

Q1

Required probability = when no. machine has fault

+ when only one machine has fault

+ when only two machines have fault.

$$= {}^5C_0 \left(\frac{3}{4}\right)^5 + {}^5C_1 \left(\frac{1}{4}\right) \left(\frac{3}{4}\right)^4 + {}^5C_2 \left(\frac{1}{4}\right)^2 \left(\frac{3}{4}\right)^3$$

$$= \frac{243}{1024} + \frac{405}{1024} + \frac{270}{1024} = \frac{918}{1024} = \frac{459}{512} = \frac{27 \times 17}{64 \times 8}$$

$$= \left(\frac{3}{4}\right)^3 \times k = \left(\frac{3}{4}\right)^3 \times \frac{17}{8}$$

$$\therefore k = \frac{17}{8}$$

Q2

A and B are independent events so

$$P\left(\frac{A}{B'}\right) = \frac{1}{3}$$

Q3

$$P(\text{exactly one}) = \frac{2}{5}$$

$$\Rightarrow P(A) + P(B) - 2P(A \cap B) = \frac{2}{5}$$

$$P(A \cup B) = \frac{1}{2}$$

$$\Rightarrow P(A) + P(B) - P(A \cap B) = \frac{1}{2}$$

$$\therefore P(A \cap B) = \frac{1}{2} - \frac{2}{5} = \frac{5-4}{10} = \frac{1}{10}$$

Q4

AA + ABA + BAA + ABBA + BBAA + BABA

$$= \frac{1}{4} + \frac{1}{8} + \frac{1}{8} + \frac{1}{16} + \frac{1}{16} + \frac{1}{16} = \frac{11}{16}$$

Q5

$$\sum p_i = 1 \Rightarrow 6k^2 + 5k = 1$$

$$6k^2 + 5k - 1 = 0$$

$$6k^2 + 6k - k - 1 = 0$$

$$(6k - 1)(k + 1) = 0 \Rightarrow k = -1 \text{ (rejected)}; k = \frac{1}{6}$$

$$P(x > 2) = k + 2k + 5k^2$$

$$= \frac{1}{6} + \frac{2}{6} + \frac{5}{36} = \frac{6 + 12 + 5}{36} = \frac{23}{36}$$

Q6

Case-I : Exactly two box contain 2, 3 balls and other two box does not contains 2, 3 balls equals to

$${}^4C_2 \times 2 \times {}^{10}C_2 \times {}^8C_3 \times \left(2^5 - \frac{5!}{2!3!} \times 2!\right)$$

$$= 2^4 \times 3^2 \times 45 \times 8 \times 7 = 2^7 \times 3^4 \times 5 \times 7$$

Case-II : Two box contains 2 balls each and two

box contain 3 balls each equals to $\frac{10!}{2!3!2!3!2!2!} \times 4!$

$$= 2^5 \times 3^3 \times 5^2 \times 7$$

$$\Rightarrow \text{probability} = \frac{2^5 \times 3^3 \times 5 \times 7 (12 + 5)}{4^{10}} = \frac{3^3 \times 5 \times 7 \times 17}{2^{15}}$$

Q7

Here $P(B_1) = P(B_2) = \frac{1}{2}$ $\{B_1, B_2 \text{ are bags}\}$ and

$$P(\text{Non-prime}) = P(B_1) \times P\left(\frac{NP}{B_1}\right) + P(B_2) \times P\left(\frac{NP}{B_2}\right) = \frac{1}{2} \times \frac{20}{30} + \frac{1}{2} \times \frac{15}{20}$$

$$\text{So, } P\left(\frac{B_1}{NP}\right) = \frac{\frac{1}{2} \times \frac{20}{30}}{\frac{1}{2} \times \frac{20}{30} + \frac{1}{2} \times \frac{15}{20}} = \frac{8}{17}$$

Q8

Here

$$P\left(\frac{E_2^C \cap E_3^C}{E_1}\right) = \frac{P[E_1 \cap (E_2^C \cap E_3^C)]}{P(E_1)}$$

$$= \frac{P(E_1) - [P(E_1 \cap E_2) + P(E_2 \cap E_3) - P(E_1 \cap E_2 \cap E_3)]}{P(E_1)}$$

$$= \frac{P(E_1) - P(E_1 \cap E_2) - P(E_2 \cap E_3) + 0}{P(E_1)}$$

$$= 1 - P(E_2) - P(E_3)$$

$$= P(E_2^C) - P(E_3) \text{ or } P(E_3^C) - P(E_2)$$

Q9

Let A is the event for getting score a multiple of 4

So, $n(A) = (1, 3), (3, 1), (2, 2), (2, 6), (6, 2), (3, 5), (5, 3), (4, 4) \& (6, 6) = 09$

$$n(A) = 9$$

$$\& n(B) = (4, 4) = 1$$

$$\text{So, } P(E) = \frac{1}{9}$$

Q10

$$\text{Sample space} = 9 \times 10^4$$

Case-I

$$\text{Out of exactly two digits selected one is zero then favourable cases} = {}^9C_1 (2^4 - 1)$$

Case-II

$$\text{Both selected digits are non-zero then favourable cases} = {}^9C_2 (2^5 - 2)$$

$$\text{Prob.} = \frac{9(2^4 - 1) + \frac{9 \cdot 8}{2}(2^5 - 2)}{9 \times 10^4}$$

$$= \frac{15 + 120}{10^4} = \frac{135}{10^4}$$

Q11

$$p = \frac{1}{10}, q = \frac{9}{10}$$

$$P(\text{not hitting target in } n \text{ trials}) = \left(\frac{9}{10}\right)^n$$

$$P(\text{at least one hit}) = 1 - \left(\frac{9}{10}\right)^n$$

$$\text{Given, } 1 - \left(\frac{9}{10}\right)^n > \frac{1}{4}$$

$$\Rightarrow (0.9)^n < 0.75$$

$$n_{\text{minimum}} = 3$$

Q12

Probability of sum getting 6 = $\frac{5}{36}$ and

Probability of sum getting 7 = $\frac{6}{36} = \frac{1}{6}$

Game ends and A wins if A throws 6 in 1st throw or A don't throw 6 in 1st throw, B don't throw 7 in 1st throw and then A throw 6 in his 2nd chance and so on.

i.e $A + \bar{A}\bar{B}A + \bar{A}\bar{B}\bar{A}\bar{B}A + \dots$

$$\Rightarrow \frac{5}{36} + \left(\frac{31}{36}\right) \left(\frac{30}{36}\right) \left(\frac{5}{36}\right) + \dots \dots \dots \infty$$

$$\Rightarrow \frac{5}{36} \left(1 + \frac{155}{216} + \left(\frac{155}{216}\right)^2 + \dots \dots \dots \infty\right)$$

$$\Rightarrow \frac{\frac{5}{36}}{\frac{61}{216}} = \frac{30}{61}$$

Q13

Probability of getting at least two 3's or 5's in one trial =

$${}^4C_2 \left(\frac{1}{3}\right)^2 \left(\frac{2}{3}\right)^2 + {}^4C_3 \left(\frac{1}{3}\right)^3 \left(\frac{2}{3}\right) + {}^4C_4 \left(\frac{1}{3}\right)^4$$

$$= \frac{33}{3^4} = \frac{11}{27}$$

$$E(x) = np = 27 \left(\frac{11}{27}\right) = 11$$

Q14

Let 'n' bombs are required, then

$$1 - {}^nC_1 \cdot \left(\frac{1}{2}\right)^1 \left(\frac{1}{2}\right)^{n-1} - {}^nC_0 \left(\frac{1}{2}\right)^0 \left(\frac{1}{2}\right)^n \geq \frac{99}{100}$$

$$\Rightarrow \frac{1}{100} \geq \frac{n+1}{2^n}$$

$$\Rightarrow 2^n \geq 100(n+1)$$

$$\Rightarrow n \geq 11$$

Q15

For an A.P. of three terms, we must select two even numbers or two odd numbers from given numbers and third number will be fixed automatically.

$$\text{Required probability} = \frac{{}^6C_2 + {}^5C_2}{{}^{11}C_3} = \frac{25}{165} = \frac{5}{33}$$

Q16

$$P(A \cap B) = P(A) + P(B) - P(A \cup B) \\ = 1 - 0.8 = 0.2$$

Now,

$$P(A \cup B \cup C) = P(A) + P(B) + P(C) - P(A \cap B) \\ - P(B \cap C) - P(C \cap A) + P(A \cap B \cap C) \\ \alpha = 0.6 + 0.4 + 0.5 - 0.2 - \beta - 0.3 + 0.2$$

$$\beta = 1.2 - \alpha$$

$$\therefore \alpha \in [0.85, 0.95]$$

$$\text{then } \beta \in [0.25, 0.35]$$

Q1 JEE Main 2020 - 7 Jan (Evening)

In a workshop there are five machines and the probability of any one of them to be out of service on a day is $\frac{1}{4}$. If the probability that at most two machines will be out of service on the same day is

$\left(\frac{3}{4}\right)^3 k$ then k is equal to:

(A) $\frac{17}{2}$

(B) 4

(C) $\frac{17}{8}$

(D) $\frac{17}{4}$

Q2 JEE Main 2020 - 8 Jan (Morning)

Let A and B be two independent events such that $P(A) = \frac{1}{3}$ and $P(B) = \frac{1}{6}$. Then, which of the following is TRUE?

(A) $P\left(\frac{A}{B}\right) = \frac{1}{6}$

(B) $P\left(\frac{A}{B'}\right) = \frac{1}{3}$

(C) $P\left(\frac{A}{B'}\right) = \frac{2}{3}$

(D) $P\left(\frac{A}{B}\right) = \frac{5}{6}$

Q3 JEE Main 2020 - 8 Jan (Evening)

Let A and B be two events such that the probability that exactly one of them occurs is $\frac{2}{5}$ and the probability that A or B occurs is $\frac{1}{2}$, then the probability of both of them occur together is

(A) $\frac{1}{10}$

(B) $\frac{2}{9}$

(C) $\frac{1}{8}$

(D) $\frac{1}{12}$

Q4 JEE Main 2020 - 9 Jan (Morning)

In a bag there are 20 cards 10 names A and another 10 names B . Cards are drawn randomly one by one with replacement then find probability that second A comes before third B .

- (A) $\frac{13}{16}$
 (B) $\frac{11}{16}$
 (C) $\frac{7}{16}$
 (D) $\frac{9}{16}$

Q5 JEE Main 2020 - 9 Jan (Evening)

A random variable X has the following probability distribution:

- $X:$ 1 2 3 4 5
 $P(x):$ K^2 $2k$ K $2K$ $5K^2$
 Then $P(x > 2)$ is equal to :

- (A) $\frac{23}{36}$
 (B) $\frac{7}{12}$
 (C) $\frac{1}{36}$
 (D) $\frac{1}{6}$

Q6 JEE Main 2020 - 9 Jan (Evening)

If 10 different balls are to be placed in 4 distinct boxes at random, then the probability that two of these boxes contain exactly 2 and 3 balls is :

- (A) $\frac{16065}{2^{15}}$
 (B) $\frac{965}{2^{10}}$
 (C) $\frac{945}{2^{10}}$
 (D) $\frac{945}{2^{11}}$

Q7 JEE Main 2020 - 2 Sep (Morning)

Box I contains 30 cards numbered 1 to 30 and Box II contains 20 cards numbered 31 to 50. A box is selected at random and a card is drawn from it. The number on the card is found to be a non-prime number. The probability that the card was drawn from Box I is :

- (A) $\frac{4}{17}$
 (B) $\frac{2}{3}$

(C) $\frac{2}{5}$

(D) $\frac{8}{17}$

Q8 JEE Main 2020 - 2 Sep (Evening)

Let E^C denote the complement of an event E . Let E_1, E_2 and E_3 be any pairwise independent events with $P(E_1) > 0$ and $P(E_1 \cap E_2 \cap E_3) = 0$

Then $P(E_2^C \cap E_3^C / E_1)$ is equal to :

(A) $P(E_3^C) - P(E_2^C)$

(B) $P(E_2^C) + P(E_3)$

(C) $P(E_3) - P(E_2^C)$

(D) $P(E_3^C) - P(E_2)$

Q9 JEE Main 2020 - 3 Sep (Morning)

A die is thrown two times and the sum of the scores appearing on the die is observed to be a multiple of 4. Then the conditional probability that the score 4 has appeared atleast once is :

(A) $\frac{1}{3}$

(B) $\frac{1}{4}$

(C) $\frac{1}{8}$

(D) $\frac{1}{9}$

Q10 JEE Main 2020 - 3 Sep (Evening)

The probability that a randomly chosen 5-digit number is made from exactly two digits is :

(A) $\frac{150}{10^4}$

(B) $\frac{134}{10^4}$

(C) $\frac{121}{10^4}$

(D) $\frac{135}{10^4}$

Q11 JEE Main 2020 - 4 Sep (Morning)

The probability of a man hitting a target is $\frac{1}{10}$. The least number of shots required, so that the probability of his hitting the target at least once is greater than $\frac{1}{4}$, is

Q12 JEE Main 2020 - 4 Sep (Evening)

In a game two players A and B take turns in throwing a pair of fair dice starting with player A and total of scores on the two dice, in each throw is noted. A wins the game if he throws a total of 6 before B throws a total of 7 and B wins the game if he throws a total of 7 before A throws a total of six. The game stops as soon as either of the players wins. The probability of A winning the game is:

- (A) $\frac{5}{31}$
- (B) $\frac{31}{61}$
- (C) $\frac{5}{6}$
- (D) $\frac{30}{61}$

Q13 JEE Main 2020 - 5 Sep (Morning)

Four fair dice are thrown independently 27 times. Then the expected number of times, at least two dice show up a three or a five, is

Q14 JEE Main 2020 - 5 Sep (Evening)

In a bombing attack, there is 50% chance that a bomb will hit the target. Atleast two independent hits are required to destroy the target completely. Then the minimum number of bombs, that must be dropped to ensure that there is at least 99% chance of completely destroying the target, is

Q15 JEE Main 2020 - 6 Sep (Morning)

Out of 11 consecutive natural numbers if three numbers are selected at random (without repetition), then the probability that they are in A.P. with positive common difference, is :

- (A) $\frac{5}{101}$
- (B) $\frac{10}{99}$
- (C) $\frac{5}{33}$
- (D) $\frac{15}{101}$

Q16 JEE Main 2020 - 6 Sep (Evening)

The probabilities of three events A , B and C are given by $P(A) = 0.6$, $P(B) = 0.4$ and $P(C) = 0.5$. If $P(A \cup B) = 0.8$, $P(A \cap C) = 0.3$, $P(A \cap B \cap C) = 0.2$, $P(B \cap C) = \beta$ and $P(A \cup B \cup C) = \alpha$, where $0.85 \leq \alpha \leq 0.95$, then β lies in the interval :

(A) $[0.25, 0.35]$

(B) $[0.35, 0.36]$

(C) $[0.36, 0.40]$

(D) $[0.20, 0.25]$

Answer Key

Q1 (C)

Q2 (B)

Q3 (A)

Q4 (B)

Q5 (A)

Q6 (A)

Q7 (D)

Q8 (D)

Q9 (D)

Q10 (D)

Q11 (3)

Q12 (D)

Q13 (11)

Q14 (11)

Q15 (C)

Q16 (A)